

Contractive Direct Limits Preserve L^p -Operator Algebras

Automated mathematics run `fa_banach_001`, lane 1

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Status. Candidate full solution, likely valid, pending expert review. For every fixed $1 \leq p < \infty$, an arbitrary contractive direct limit of L^p -operator algebras is again an L^p -operator algebra. No complete-contractivity or injectivity hypothesis on the connecting maps is required.

1 Source question

Phillips [1] defines an L^p -operator algebra to be a Banach algebra isometrically isomorphic to a norm-closed subalgebra of $\mathcal{B}(L^p(X, \mu))$. Proposition 6.1 of that paper constructs the direct limit $A = \varinjlim A_i$ of any directed system of Banach algebras and contractive homomorphisms. If $\varphi_i : A_i \rightarrow A$ are the canonical maps, it proves

$$\|\varphi_i(a)\| = \lim_{j \geq i} \|\varphi_{j,i}(a)\|, \quad a \in A_i, \quad (1)$$

and that $\bigcup_i \varphi_i(A_i)$ is dense in A .

Immediately afterward, on PDF page 35, the paper asks whether A is an L^p -operator algebra when every A_i is one.

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If the A_i in Proposition 6.1 are L^p operator algebras, we do not know whether it follows that $\varinjlim A_i$ is an L^p operator algebra.

Corollary 6.2. Let A be a Banach algebra, let I be a directed set, and let $(A_i)_{i \in I}$ be a family of closed subalgebras of A such that $A_i \subset A_j$ for $i \leq j$, and such that $\bigcup_{i \in I} A_i = A$. For $i \leq j$, let $\varphi_{j,i} : A_i \rightarrow A_j$ be the inclusion. Then the canonical map from $\varinjlim A_i$ to A is an isometric isomorphism.

Proof. The formula for $\|\varphi_i(a)\|$ in Proposition 6.1 implies that the map $\varinjlim A_i \rightarrow A$ is isometric. Therefore density of $\bigcup_{i \in I} A_i$ in A implies surjectivity. $\square$

Figure 1: The source question and its immediate context on page 35 of Phillips [1].

Throughout the result below, $1 \leq p < \infty$, the range of the source paper.

2 Proof intuition

Represent each stage A_j isometrically on some L^p -space E_j . A stage element $a \in A_i$ then has a coherent tail of represented images $\rho_j(\varphi_{j,i}(a))$ for $j \geq i$. Choose an ultrafilter which regards every

final segment as large and let this operator tail act on the Banach-space ultraproduct $(E_j)_\mathcal{U}$.

There are two exact norm facts. First, the norm of an operator induced by a bounded family is the ultralimit of the coordinate operator norms. Second, the ordinary tail limit of those norms is precisely the direct-limit norm in (1). Thus the tail construction is isometric. It is also well defined when two stage elements become equal only asymptotically, since then the tail norm of their difference tends to zero. The dense algebraic limit consequently embeds isometrically, and the embedding extends to its completion. Finally, ultraproducts of L^p -spaces are L^p -spaces for finite p .

3 The theorem

We first record the elementary operator-ultraproduct norm identity used in the proof.

Lemma 1. *Let $(E_j)_{j \in I}$ be nonzero Banach spaces, let $\mathcal{U}$ be an ultrafilter on I , and let $(T_j)_{j \in I}$ be a uniformly bounded family with $T_j \in \mathcal{B}(E_j)$. The coordinatewise operator*

$$T_\mathcal{U} : (E_j)_\mathcal{U} \longrightarrow (E_j)_\mathcal{U}, \quad [(x_j)] \longmapsto [(T_j x_j)],$$

satisfies

$$\|T_\mathcal{U}\| = \lim_{j \rightarrow \mathcal{U}} \|T_j\|.$$

Proof. The upper bound follows directly from the ultraproduct norm. Fix $\varepsilon > 0$. For every j , choose $x_j \in E_j$ with $\|x_j\| = 1$ and

$$\|T_j x_j\| \geq \|T_j\| - \varepsilon.$$

Then $x = [(x_j)]$ has norm one and

$$\|T_\mathcal{U} x\| = \lim_{j \rightarrow \mathcal{U}} \|T_j x_j\| \geq \lim_{j \rightarrow \mathcal{U}} \|T_j\| - \varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves the reverse bound. □

Theorem 1. *Fix $1 \leq p < \infty$. Let I be a directed set and*

$$((A_i)_{i \in I}, (\varphi_{j,i})_{i \leq j})$$

be a direct system of L^p -operator algebras whose connecting homomorphisms are contractive. Then its direct limit in the category of Banach algebras and contractive homomorphisms is an L^p -operator algebra.

Proof. For $i \in I$, put $I_i = \{j \in I : j \geq i\}$. The family of final segments $(I_i)_{i \in I}$ has the finite-intersection property, so choose an ultrafilter $\mathcal{U}$ on I which contains every I_i .

For each $j \in I$, choose an isometric algebra homomorphism

$$\rho_j : A_j \longrightarrow \mathcal{B}(E_j), \quad E_j = L^p(X_j, \mu_j),$$

with $E_j \neq \{0\}$. (If necessary, take the p -direct sum of a given representation with the zero representation on $\mathbb{C}$.) Let $E = (E_j)_\mathcal{U}$. The standard L^p ultraproduct theorem says that E is isometrically an L^p -space; see Heinrich [2].

For $i \in I$ and $a \in A_i$, define a uniformly bounded operator family by

$$T_j^{i,a} = \begin{cases} \rho_j(\varphi_{j,i}(a)), & j \geq i, \\ 0, & j \not\geq i. \end{cases}$$

Contractivity gives $\sup_j \|T_j^{i,a}\| \leq \|a\|$. Let $T^{i,a} \in \mathcal{B}(E)$ be its induced ultraproduct operator.

We use this to define an operator on the dense subalgebra

$$A_0 = \bigcup_{i \in I} \varphi_i(A_i) \subset A$$

by setting $\rho(\varphi_i(a)) = T^{i,a}$. We first check that this is well defined. Suppose $\varphi_i(a) = \varphi_k(b)$ in A . Choose $\ell \geq i, k$ and set

$$d = \varphi_{\ell,i}(a) - \varphi_{\ell,k}(b) \in A_\ell.$$

By (1),

$$\lim_{j \geq \ell} \|\varphi_{j,\ell}(d)\| = \|\varphi_\ell(d)\| = 0.$$

On the final segment I_ℓ , the difference between the two operator families is $\rho_j(\varphi_{j,\ell}(d))$. Lemma 1 therefore gives $T^{i,a} = T^{k,b}$.

Linearity and multiplicativity are checked on a common final segment. For example, if $\ell \geq i, k$, then for every $j \geq \ell$,

$$\begin{aligned} T_j^{i,a} T_j^{k,b} &= \rho_j(\varphi_{j,i}(a) \varphi_{j,k}(b)) \\ &= \rho_j(\varphi_{j,\ell}(\varphi_{\ell,i}(a) \varphi_{\ell,k}(b))) , \end{aligned}$$

which is the tail family representing the product in A_0 .

Finally, Lemma 1, the isometry of each ρ_j , and (1) give

$$\begin{aligned} \|\rho(\varphi_i(a))\| &= \lim_{j \rightarrow \mathcal{U}} \|T_j^{i,a}\| \\ &= \lim_{j \rightarrow \mathcal{U}} \|\varphi_{j,i}(a)\| \\ &= \lim_{j \geq i} \|\varphi_{j,i}(a)\| \\ &= \|\varphi_i(a)\|. \end{aligned}$$

Thus $\rho : A_0 \rightarrow \mathcal{B}(E)$ is an isometric algebra homomorphism. Since A_0 is dense in A , it extends uniquely to an isometric homomorphism $A \rightarrow \mathcal{B}(E)$. Its range is closed because A is complete. Since E is an L^p -space, A is an L^p -operator algebra. $\square$

4 Scope, prior art, and novelty check

Theorem 1 answers the page-35 question throughout the finite- p range of the source paper. It does not assume that the connecting maps are injective, isometric, nondegenerate, or completely contractive, and the index set need not be countable. No claim is made for $p = \infty$.

Phillips's 2014 GPOTS talk summary [3, Theorem 4.1] announces that a direct limit of L^p -operator algebras with *completely contractive* maps is an L^p -operator algebra and mentions the same ultraproduct mechanism. The source problem, however, assumes only ordinary contractivity. The proof above uses neither matrix norms nor amplifications, and so removes that additional hypothesis.

The local run indexes were searched for arXiv:1309.6406, the exact title, “direct limit,” “ L^p -operator algebra,” and “ultraproduct.” Bounded web searches on August 9, 2026 used the exact source sentence and combinations of the author, title, contractive/completely contractive maps, direct limits, and ultraproducts. They found the source, the 2014 talk's completely-contractive

special case, and later special direct-limit constructions, but no treatment of arbitrary contractive connecting homomorphisms. Novelty confidence is therefore positive but conservative pending expert literature review.

No computation is used. The proof audit in `VERIFICATION.md` identifies the well-definedness and norm-preservation steps for focused human review.

References

- [1] N. C. Phillips, *Crossed products of L^p operator algebras and the K -theory of Cuntz algebras on L^p spaces*, arXiv:1309.6406 (2013).
- [2] S. Heinrich, *Ultraproducts in Banach space theory*, J. Reine Angew. Math. **313** (1980), 72–104; doi:10.1515/crll.1980.313.72.
- [3] N. C. Phillips, *Operator algebras on L^p spaces which “look like” C^* -algebras*, GPOTS talk summary (2014), Theorem 4.1.